COMPUTER SCIENCE –LEVEL 6
UNDERSTAND COMPUTER ORGANIZATION AND ARCHITECTURE
CT/CU/CS/CR/01/6/8
1 1/2 HOURS
JULY 2025

JKUAT TVET INSTITUTE – KAREN CAMPUS
WRITTEN ASSESSMENT

CAT 1

Answer all the questions.

SECTION A (20 marks)

1. Define the following terms as related to computer organisation
   a. Computer memory unit
   b. Computer architecture (4 marks)
2. Memory is categorised into either volatile or non-volatile. Distinguish between volatile memory and non-volatile memory. (4 marks)
3. Cache memory is an essential part of the CPU, ensuring its best performance. Describe the cache memory works (4 marks)
4. Processors are mainly categorised based on the task they perform. Explain **the two** types of processors. (4 marks)
5. In order to store data permanently, computers rely on secondary storage devices. List **any four** types of secondary memory (4 marks)

SECTION B (30 marks)

6. The Central Processing Unit has its main function as the brain of the computer.
   a. Describe the **three** main components of the CPU (6 marks)
   b. List **any four** key factors about CPU architecture that affect its performance (4 marks)
7. A computer is organised into several components that work together to perform efficiently. Describe any **FIVE** components of a computer system (10 marks)
8. .
   a. Your computer's performance heavily relies on how efficiently memory is managed. Explain **any two** types of RAM (4 marks)
   b. You are planning to upgrade the RAM on your personal computer to enhance its speed and multitasking capabilities. Explain **any three** key specification factors that should be considered when selecting RAM for your computer (6 marks)

CAT 1

40/50

JKUAT CONTINOUS ASSESSEMENT TESTS

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REG NO: TOC 113-C002-0063/2024

COURSE: Computer Science YEAR OF STUDY: 1

SHEET NO:

UNIT CODE: TITLE: Computer Architecture

DATE: 3/7/2025

NOTE: This stationery will be used for Continuous Assessment work only

It will be a breach of examination regulations to use it otherwise.

1.a) Compute Memory unit - is a collection (2) of storage devices that work together to store data temporarily or permanently.

b) Computer architecture - Is ~~a collection of~~ the internal working of ~~components~~ components within a system to form a computer system that is the CPU, main memory and the control unit. (2)

2. Volatile memory is a memory whereby it easily looses it's data when the power is off while non-volatile memory is a memory ~~is~~ whereby it retains it data even if the power is off. (2) (2)

3. Cache memory is a memory which has the capability to store frequently used data. It is small in size and compact to execute and store a lot of data? (1)

4. RISC - This is a Reduced Instruction Set Computer processor which execute small set amount of highly optimized data while CISC which is Complex Instruction Set Computer it's function is to perform complex operations at a given set of instructions. (2) (2)

5. i) Optical discs (1)
ii) Flash memory (1)
iii) Hard Disk Drive (1)
iv) Solid State Drive (1)

SECTION B

6 a) Control Unit
- This is a CPU component (2) whose main function is to command how instructions should be done in other term it acts as the manager.

ALU (Arithmetic Logic Unit)
- This is a CPU component (2) whose function is to perform arithmetic operations and logic operations. The arithmetic operations include addition, substraction while logic operations include 'NOT', 'AND' and 'OR'

Main Memory
- This CPU component (2) whose function is store data which is likely to be used right now inorder to be executed.

b) Clock Speed - This is the rate at which data is processed at a given cycle and it is measured in MegaHertz or GigaHertz.
- Core

b) List any key factors about CPU architecture (1)
i) Core (1)
ii) Clock Speed (1)
iii) Type of the processor
iv) Voltage

~~7) CPU~~ -

7 i) CPU - is a component ~~of~~ whose main function acts as the brain of the computer (2)

ii) Output Devices - is a component whose main function is to produce the results of the computer work. eg Monitor, Printer. (1)

iii) Input Device - is a component whose main function is to ~~cent~~ control things like typing data and executing commands. eg ~~Mon~~ Mouse, Keyboard. (1)

iv) Main Memory - is a component whose main function is to store data that is going to be ~~used~~ executed right now. (2)

v) RAM - is a component whose main function is to store data temporarily

8a) Static RAM - whose function is storing data using the capacitors. (2)
Dynamic RAM - whose function is storing data using the flip flop.

b) Latency - This is how the RAM responds to request hence lower latency affects the ~~per~~ computer performance of the RAM. (2)
Speed - This is the rate at which data is processed in a unit ~~of units~~ measured in MegaHertz ~~or~~ GigaHertz. (2)
Voltage - This is the rate at which the RAM consumes power in volts hence lower voltage can affect the performance of the RAM due to low consumption of power. (2)